Adamya Kumar

Software Development Engineer | Full Stack Developer | MERN Stack
Bangalore, India | +91-7310839922 | adamyakumar2307@gmail.com
linkedin.com/in/adamya-kumar | github.com/Adamya-Kumar | portfolio-website-ai-src.vercel.app

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, SQL, HTML, CSS
Frontend: React, Next.js, Vite, Tailwind CSS, TanStack Query, React Router
Backend: Node.js, Express.js, FastAPI, RESTful APIs, Microservices, JWT Authentication, RBAC, Zod, SSE
Databases: MongoDB, Mongoose, PostgreSQL, MySQL, ChromaDB, Pinecone, FAISS
DevOps and Tools: Git, GitHub, Docker, CI/CD, AWS, Vercel, Render, Streamlit Cloud, Cloudinary, VS Code, Jupyter
AI Development Tools: GitHub Copilot, Claude, Cursor, Spec-Driven Development, Prompt Engineering
GenAI and Agents: LangChain, LangGraph, RAG, OpenAI APIs, Google Gemini, Groq, Multi-Agent Systems, Tool Calling
Computer Science: Data Structures and Algorithms, Object-Oriented Programming (OOP), Problem Solving, Testing

PROJECTS

Spark Commerce, Full Stack MERN E-Commerce Platform | *React*, *Node.js*, *Express.js*, *MongoDB*, *Tailwind CSS*, *Vercel*
Live: www.carteer.space

- Built a full stack multi-category storefront and admin panel with React (Vite), React Router, TanStack Query, and Tailwind CSS: product browsing, cart, Cash on Delivery checkout, and order management; deployed on Vercel with a custom domain.
- Implemented httpOnly-cookie JWT authentication with customer and admin roles on a Node.js and Express RESTful API with Mongoose models, and validated requests with Zod.
- Kept pricing on the server: money stored as integer paise, with shipping rules (free above Rs. 999, Rs. 49 otherwise) applied in the backend and never trusted from the client.
- Structured the system as three microservice-style services (React client, Express API, Python assistant) with clear ownership boundaries, so the AI layer can change without touching business rules.
- Added admin catalog management with Cloudinary image uploads; designed a FastAPI and LangGraph shopping assistant (Gemini, FAISS) that the Express API proxies over SSE.
- Built unit by unit with spec-driven development, using Cursor and GitHub Copilot as AI coding assistants and reviewing the generated code before merging.

Production RAG Bot | *Python*, *LangChain*, *ChromaDB*, *Groq*, *Google Gemini*, *Hugging Face*, *Streamlit*
Live: pro-rag-chatbot.streamlit.app

- Built a RAG chatbot where users upload multiple PDFs and ask questions about them; handles PDF parsing (pypdf), chunking, embeddings, and retrieval-based answers with LangChain.
- Supported Groq (Llama 3.1 and Llama 3 70B) and Google Gemini (2.0 and 2.5 Flash) with provider-specific embeddings (sentence-transformers for Groq, Google embeddings for Gemini), so users can switch models without starting over.
- Persisted vectors in ChromaDB with automatic reuse and append-on-upload, so documents are not re-processed.
- Organized the code into modules (chat, sidebar, LLM, vector store, PDF handling) and added chat history export, clear/reset/undo controls, and a developer view to inspect stored chunks; deployed on Streamlit Cloud with session-based chat state.

ResearchMind, Multi-Agent AI Research System | *Python*, *LangChain*, *Streamlit*, *Docker*, *Tavily*, *OpenRouter*
Live: researchmind-multi-agent-ai-research.onrender.com

- Built a four-stage agent pipeline: a Search agent (Tavily) finds recent sources, a Reader agent scrapes them with BeautifulSoup, a Writer chain drafts the report, and a Critic chain reviews and scores it.
- Created a Streamlit interface that shows each agent stage in real time and lets users download the final report as Markdown.
- Containerized with Docker and deployed on Render; kept API keys in environment variables with an .env.example template.

EDUCATION

Bachelor of Technology in Computer Science and Engineering Sep. 2021 – Jun. 2025
Graphic Era Hill University, Dehradun, India

ACHIEVEMENTS AND CERTIFICATIONS

- Top 10 Contributor, Hacktoberfest 2023 (among 2,000+ developers); solved 250+ DSA problems on LeetCode and Codeforces.
- Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms (Coursera, DeepLearning.AI).
- Spring Boot: Microservices, Kafka, Docker, Kubernetes, AWS (CodingShuttle).